# Unit ID: 679b01f9aaf593b2823f2dbf

# Unit 2 - Data and Information Processing

## PDF Documents

### PDF Document 1: 1738211831052-Chapter 1 - Unit 2 - Data and Information Processing.pdf

This PDF document is included in the unit materials.

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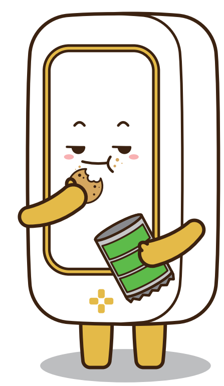
# UNIT 2 DATA AND INFORMATION PROCESSING

## What is Data?

Data is any information a computer processes, such as numbers, letters, or images. When you type a message, play a game, or save a photo, you're creating data. But how does a computer understand this data? Computers use a unique language called binary code, which comprises only two numbers: 0 and 1.

Example: The letter “A” in binary looks like this: 01000001. Computers read these 0s and 1s and turn them into letters, images, or sounds we can understand.



## WHY NETWORKS AND THE INTERNET ARE IMPORTANT

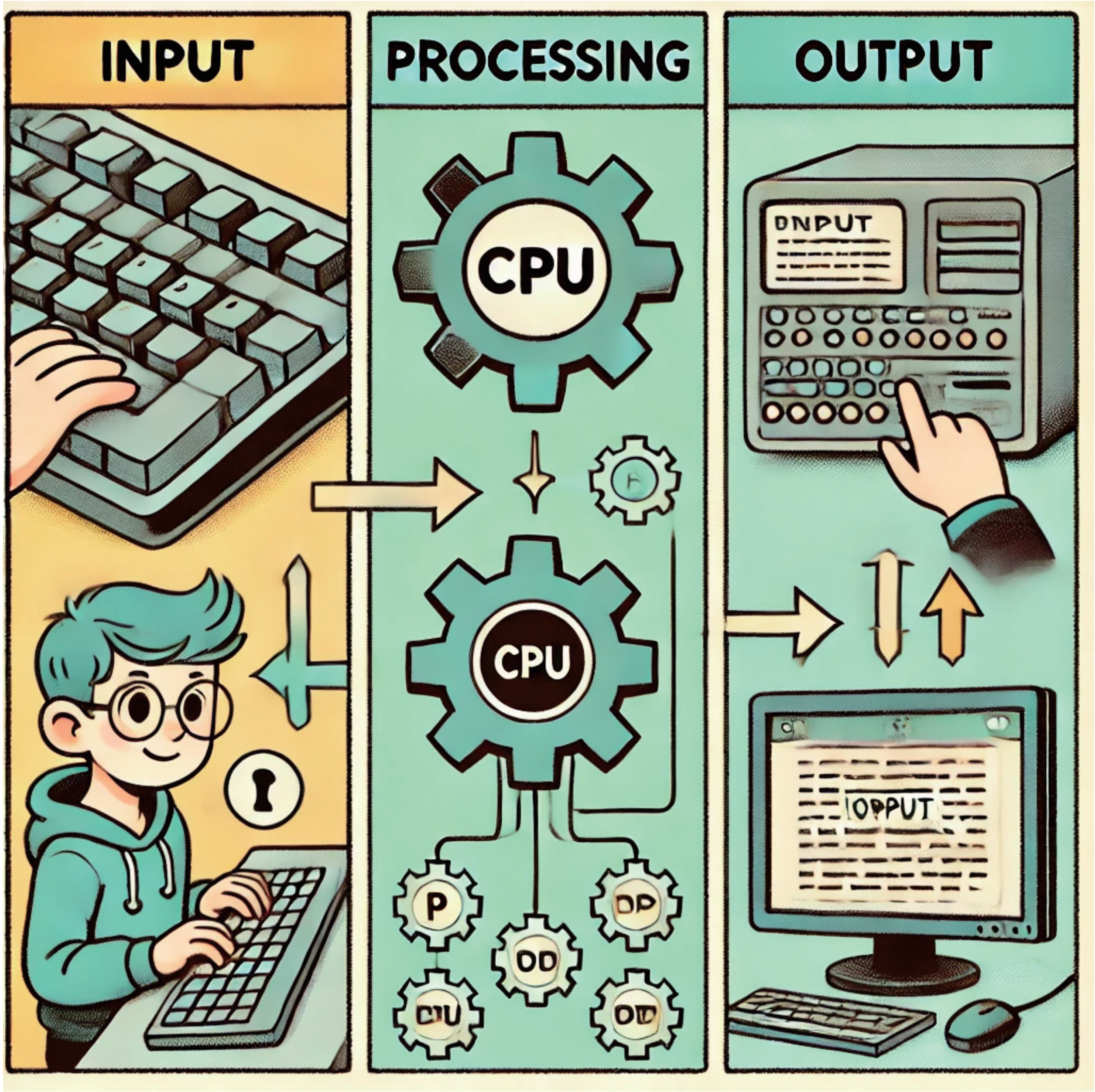
When a computer processes data, it turns it into useful information. For example:
Typing on a Keyboard: Each key you press sends data to the computer, processes it and displays the letter on the screen.

### EXAMPLES OF DATA PROCESSING:

- **Typing on a Keyboard:** Data is sent to the computer whenever you press a key on your keyboard. This data is processed, resulting in the letter or symbol appearing on your screen. The computer processes your keystrokes in realtime to create words, sentences, and documents.
- **Photos and Videos:** A camera captures light and colors as raw data. This data includes many tiny bits of information about the brightness and color of each pixel in the image. The computer processes this data into a complete photo or video that you can see and share.

## THE PROCESS OF DATA FLOW

Data doesn't just appear on your screen by magic—it follows a specific path inside the computer. This path is divided into three main stages that help the computer take data and turn it into information:




**1 Input:** This is the stage where data enters the computer. Examples of input include:

- Typing on a keyboard to write a message.
- Clicking with a mouse to select an option or move a cursor.
- Speaking into a microphone for voice commands or recording audio.
- Capturing a photo or video using a camera.

Input devices like keyboards, mice, microphones, and cameras collect this raw data and send it to the computer for processing.

**2** Processing: Once the data has entered the computer, the processor (the CPU or central processing unit) takes over. The CPU is like the computer's brain; it follows specific instructions to analyze and organize the data. For example:

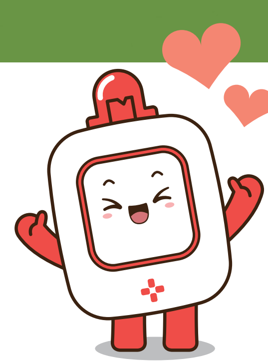
- When you type, the CPU decodes each key press and arranges the letters into sentences on your screen.
- When a photo is captured, the CPU processes the light and color data to produce a highquality image.

During this stage, the computer uses various software programs and algorithms to ensure the data is processed efficiently and accurately.

**3** Output: After processing the data, the computer produces output. This output is what you see, hear, or interact with:

- Text on the screen when you type or search for something.
- Images and videos displayed on the monitor after being processed.
- Sound through speakers when you play a song or a video.

Output devices like monitors, speakers, and printers help show or play the processed information, making it useful for you.



## Activity: Create a Data Flow Chart

Draw a simple flowchart showing how data moves when you search for a picture online. Include input (typing the search), processing (computer finding the picture), and output (showing the image on the screen).